

Mathematical Model to Assess the Relative Effectiveness of Rift Valley Fever Countermeasures

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ABSTRACT

Mathematical modeling of infectious diseases is increasingly used to explicate the mechanics of disease propagation, impact of controls, and sensitivity of countermeasures. The authors demonstrate use of a Rift Valley Fever (RVF) model to study efficacy of countermeasures to disease transmission parameters. RVF is a viral infectious disease that propagates through infected mosquitoes and primarily affects animals but also humans. Vaccines exist to protect against the disease but there is lack of data comparing efficacy of vaccination with alternative countermeasures such as managing mosquito population or destroying infected livestock. This paper presents a compartmentalized multispecies deterministic ordinary differential equation model of RVF propagation among livestock through infected Aedes and Culex mosquitoes and exercises the model to study the efficacy of vector adulticide, vector larvicide, livestock vaccination, and livestock culling on livestock population. Results suggest that livestock vaccination and culling offer the greatest benefit in terms of reducing livestock morbidity and mortality.

Keywords: *Disease Modeling, Disease Transmission, Mathematical Modeling, Rift Valley Fever, Vaccination Effectiveness*

INTRODUCTION

Rift Valley fever virus is a mosquito-borne pathogen that causes widespread febrile illness and mortality in domestic animals such as

sheep, cattle and goats as well as humans (Gaff, 2007). Rift Valley fever virus was first described in peer-reviewed research in 1930 (Daubney, 1931) and was generally considered a disease primarily of sub-Saharan and southern Africa (Gaff, 2007). Then in 1977, the disease moved outside of sub-Saharan and southern Africa with

DOI: 10.4018/ijalr.2011040101

an outbreak occurring in Egypt; since then, outbreaks have occurred in Saudi Arabia and Yemen proving it to be a virus able to invade ecologically diverse regions (Gaff, 2007).

Over the past few decades, significant changes in the distribution and intensity of Rift Valley fever (RVF) have been recorded (WHO, 2007). Since the isolation of the virus in 1931 in the Rift Valley in Kenya, it has been held responsible for several epizootics in small ruminants, causing abortions and stillborns in the ovine species in Eastern and Southern Africa (Gerdes, 2004). Epizootics first occurred in regions of high altitude such as South Africa in 1951 (which resulted in the death of an estimated 100,000 sheep), Zimbabwe in 1958, Nigeria in 1958, and Chad and Cameroon in 1967 (WHO, 2007). Until the 1970s human infection remained low, and the agent mostly affected breeders in contact with affected or dead animals. In 1973, after the first source of infection appeared in the White Nile in Sudan, a human epidemic soon began in South Africa with the first recognized human deaths (Peters, 1994). Human outbreaks then occurred in Egypt in 1977 causing 598 human deaths (Gerdes, 2004), in 1987 causing 200 human deaths, and in Kenya and Somalia in 1997 causing 478 human deaths (CDC, 1998). In 2000, cases of RVF were discovered in Saudi Arabia and Yemen marking the spread of the disease outside of Africa and the Rift Valley (Jupp, 2002). By November 2000, over 500 cases of serious RVF were discovered in Saudi Arabia, with 87 deaths. In Yemen, between August and November 2000, there were over 1,000 suspected occurrences of the disease among humans. The result of the outbreak in Yemen was 121 deaths. Since 2000, outbreaks have occurred in Kenya and Somalia (2006), Tanzania and Sudan (2007) and Madagascar and South Africa (2008).

The potential for an exotic arbovirus to be introduced and widely established across North America can be inferred by the introduction and rapid spread of West Nile viral activity across

North America in 1999 (Turell, 2008). Currently, Rift Valley fever is listed as a Category A agent on the Center for Disease Control bioterrorism list and is therefore considered a major threat to the United States. Despite the existence of several vaccines, which can protect against RVF, livestock vaccination is not currently a standardized activity in the United States. This leaves livestock culling of both exposed and infected animals as the only viable after-the-fact countermeasures. The disease is spread by infected mosquitoes whose population can vary widely, and such variance is hard to control as it primarily depends on environment factors. Should RVF be intentionally or unintentionally introduced in the Continental United States, exposed livestock is likely to encompass a significant percentage of all livestock near the vicinity of first appearance. Under such a scenario, the potential economic disruption due to loss of livestock and any subsequent trade restrictions is significant.

Disease control and public health intervention measures for RVF entail three primary countermeasures: the application of insecticide to vector populations targeting either adult mosquitoes or mosquito larvae, livestock vaccination, or finally culling of exposed and/or infected animals. Each of these countermeasures has an associated a-priori cost and corresponding posterior effectiveness; however, there is little research that allows a quantitative cost-benefit analysis among these countermeasures. In this paper, we extend a mathematical model of RVF to include transmission of RVF between two vector populations and a single livestock population. The resultant model allows experimentation with four specific intervention strategies: vector adulticide, vector larvicide, livestock vaccination, and livestock culling. We utilize the enhanced model to explore each of these systems for various degrees of intensity and efficacy of intervention, and to evaluate the sensitivity of the models to the various disease transmission parameters.

METHODS

The RVF Model

We extend the mathematical model for Rift Valley fever originally presented in (Gaff, 2007). The model itself is a compartmentalized multi-species deterministic ordinary differential equation system, which predicts the spread of Rift Valley Fever in *Aedes* and *Culex* mosquitoes and a livestock population (in this case, cattle). The model is based on a standard framework for epidemiology models using disease states as the population compartments: SEIR. S are the susceptible individuals of a given population with no assumed immunity or previous exposure to the disease. E is the exposed individuals who have been exposed and infected but are not yet infectious. I is the infectious individuals who can now expose others and have symptoms. R are the recovered individuals who are recovered and assumed immune from the disease. In addition to this framework, the mosquito populations have larval compartments, P. The infected *Aedes* mosquitoes are capable of laying infected eggs so there is an additional larval compartment, Q. Movement between compartments for the SEIR model is determined by disease dynamics while age controls flow between the larval compartments (PQ) and the appropriate adult compartment (S or I). Subscripts are used to identify the three populations: (1) *Aedes*, (2) *Culex*, (3) livestock.

The original model has been expanded to accommodate vector control, livestock vaccination and culling. The model has been modified to include logistic growth for the mosquito populations as replacement birth and death does not make sense when modeling vector control measures. The carrying capacity for the populations will eventually be tied to climate factors such as rainfall. The four modified models, one with each type of intervention, are represented schematically in Figures 1(a) through (d); the full set of equations for the general intervention model is provided in Table 1, and represents all interventions simultaneously: adulticide ($v_A(t)$), larvicide ($v_L(t)$), livestock vaccination ($V(t)$) and livestock culling ($c(t)$).

Model Parameters and Initial Values

Vector and livestock parameters were set based upon species biology and are shown in Table 2. Also shown are disease parameters, which were based on the behavior of RVF in mosquitoes and cattle, and high and low transmission parameters, which were derived previously for the development of the original model. The simulation was run over a 10-year period using Runge-Kutta 4th order methods to solve the differential equations, which have a one-day time step.

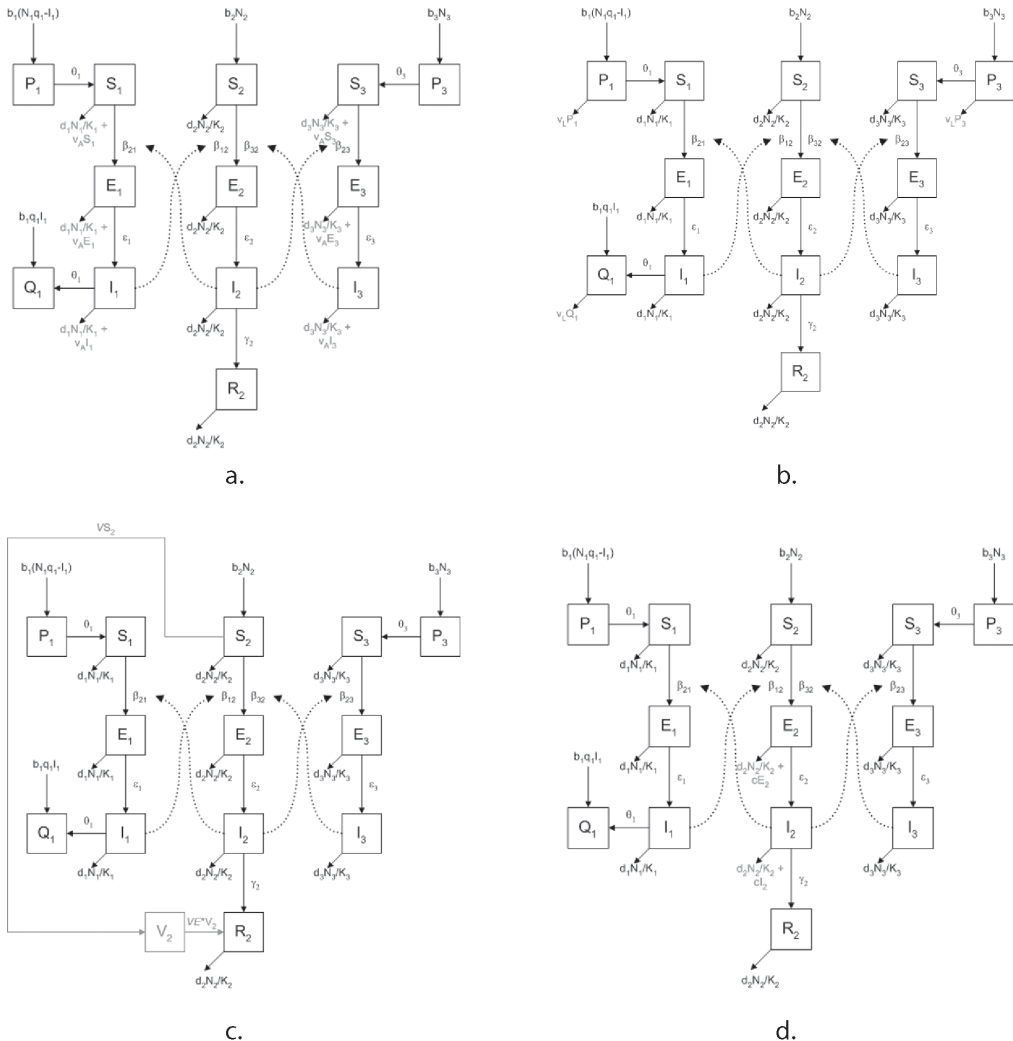
For the analysis performed, each intervention was explored singly. Simulations in which vector control and culling are employed, as well as reactive livestock vaccination (defined below), are initiated with all livestock individuals and adult *Culex* mosquitoes in the susceptible compartment, and with the majority of *Aedes* mosquitoes in the susceptible compartment and a single infected *Aedes* adult (Table 3). Simulations employing preventive livestock vaccination are initiated with a given proportion of the livestock population already in the removed state, indicating pre-existing protection against RVF prior to the outbreak of disease in the system (Table 4).

The initial conditions for livestock population numbers under preventive vaccination scenarios were obtained by implementing given combinations of target population and vaccine efficacy values in the absence of disease transmission and allowing the system to reach equilibrium. The resulting proportion of immune livestock ($PROP$) is then employed as the initial proportion of immune for simulations in which a single *Aedes* mosquito is infected, and the initial proportion of susceptible livestock in these runs is defined as $1-PROP$.

Vector Control Interventions

Vector control is incorporated in the model by proportionally reducing the population numbers for adult and larva vector populations by 30%, 60% and 90% for both adulticide ($v_A(t)$) and larvicide ($v_L(t)$). In the case of adulticide,

Figure 1. Flow diagram for the model of Rift Valley Fever transmission. Note that red denotes the key differences for each of the models. (a) Vector adulticide intervention. (b) Vector larvicide intervention. (c) Livestock vaccination intervention. (d) Livestock culling intervention.



all adult mosquitoes in the treated populations are affected equally, regardless of current disease state; similarly, in the case of larvicide, all mosquito larvae are impacted uniformly. The application of insecticide is repeated four times on a weekly basis for periodic scenarios, occurs a single time for one-time scenarios, or continues for 28 consecutive days for continuous scenarios. Reactive vector control is initiated

once thresholds for a detectable epidemic have been reached in the simulation, defined as at least 1% disease prevalence in the livestock population or two livestock deaths attributable to RVF in a given time period. Preventive vector control is initiated at the beginning of simulations, and prior to any detectable disease in livestock, though with initial conditions including a single infected adult *Aedes* mosquito.

Table 1. Equations for the model of Rift Valley Fever transmission. Subscript 1 denotes *Aedes* mosquitoes, subscript 2 denotes *Culex* mosquitoes, and subscript 3 denotes livestock.

$\frac{dP_1}{dt} = b_1(N_1 - q_1I) - \theta_1P_1 - v_L(t)P_1$ $\frac{dQ_1}{dt} = b_1q_1I_1 - \theta_1Q_1 - v_L(t)Q_1$ $\frac{dS_1}{dt} = \theta_1P_1 - \frac{d_1S_1N_1}{K_1} - \frac{\beta_{21}S_1I_2}{N_2} - v_A(t)S_1$ $\frac{dE_1}{dt} = -\frac{d_1E_1N_1}{K_1} - \frac{\beta_{21}S_1I_2}{N_2} - \varepsilon_1E_1 - v_L(t)E_1$ $\frac{dI_1}{dt} = \theta_1Q_1 - \frac{d_1I_1N_1}{K_1} + \varepsilon_1E_1 - v_A(t)I_1$ $\frac{dN_1}{dt} = (\theta_1 - v_L(t))(P_1 + Q_1) - \frac{d_1N_1N_1}{K_1} - v_A(t)N_1$	$\frac{dP_3}{dt} = b_3N_3 - \theta_3P_3 - v_L(t)P_3$ $\frac{dS_3}{dt} = \theta_3P_3 - \frac{d_3S_3N_3}{K_3} - \frac{\beta_{23}S_3I_2}{N_2} - v_A(t)S_3$ $\frac{dE_3}{dt} = -\frac{d_3E_3N_3}{K_3} - \frac{\beta_{23}S_3I_2}{N_2} - \varepsilon_3E_3 - v_L(t)E_3$ $\frac{dI_3}{dt} = -\frac{d_3I_3N_3}{K_3} + \varepsilon_3E_3 - v_A(t)I_3$ $\frac{dN_3}{dt} = (\theta_3 - v_L(t))P_3 - \frac{d_3N_3N_3}{K_3} - v_A(t)N_3$
$\frac{dS_2}{dt} = b_2N_2 - \frac{d_2S_2N_2}{K_2} - \frac{\beta_{12}S_2I_1}{N_1} - \frac{\beta_{32}S_2I_3}{N_3} - V(t)S_2$ $\frac{dE_2}{dt} = -\frac{d_2E_2N_2}{K_2} + \frac{\beta_{12}S_2I_1}{N_1} + \frac{\beta_{32}S_2I_3}{N_3} - \varepsilon_2E_2 - c(t)E_2$ $\frac{dI_2}{dt} = -\frac{d_2I_2N_2}{K_2} + \varepsilon_2E_2 - \gamma_2I_2 - \mu_2I_2 - c(t)I_2$ $\frac{dV_2}{dt} = V(t)S_2 - VE * V_2$ $\frac{dR_2}{dt} = -\frac{d_2R_2N_2}{K_2} + \gamma_2I_2 + VE * V_2$ $\frac{dN_2}{dt} = b_2N_2 - \frac{d_2N_2N_2}{K_2} - \mu_2I_2 - c(t)(E_2 + I_2)$	

Table 5 summarizes the vector control scenarios that have been simulated.

Livestock Vaccination Interventions

Livestock vaccination is incorporated in the RVF model by allowing all of the adult livestock population to receive a single immunization against the disease. Once vaccinated, a susceptible animal will be protected according to the specific vaccine efficacy (modeled here to be 65% or 85%) after a period of 10 days elapses; though all livestock are vaccinated, only

susceptible animals receive benefit from it and all others remain in their current disease state.

As with vector control, livestock vaccination scenarios occur on a periodic, one-time or continuous basis, with timing definitions carrying over. Reactive vaccination occurs either in response to a detectable livestock epidemic as defined above, or in response to the detection of disease in a single mosquito. Preventive vaccination occurs prior to the introduction of RVF into the model system, and has the effect of altering the initial conditions for livestock such that a proportion of the animal population shifts from the susceptible disease state into the

Table 2. Parameter values for the model of Rift Valley Fever transmission

Parameter	Description	Unit	Value	High Transmission	Low Transmission
β_{12}	Adequate contact rate from <i>Aedes</i> to livestock	1/ day		0.480	0.150
β_{21}	Adequate contact rate from livestock to <i>Aedes</i>	1/ day		0.395	0.150
β_{23}	Adequate contact rate from livestock to <i>Culex</i>	1/ day		0.560	0.150
β_{32}	Adequate contact rate from <i>Culex</i> to livestock	1/ day		0.130	0.050
μ_2	RVF mortality rate in livestock (cattle)	1/ day	0.0176		
$1/d_1$	Lifespan of <i>Aedes</i> mosquitoes	days	10		
$1/d_2$	Lifespan of livestock	days	3650		
$1/d_3$	Lifespan of <i>Culex</i> mosquitoes	days	10		
b_1	Number of <i>Aedes</i> eggs laid per day	1/ day	d_1		
b_2	Daily birthrate of livestock	1/ day	d_2		
b_3	Number of <i>Culex</i> eggs laid per day	1/ day	d_3		
$1/\varepsilon_1$	Disease incubation period in <i>Aedes</i> mosquitoes	days	6		
$1/\varepsilon_1$	Disease incubation period in livestock	days	4		
$1/\varepsilon_1$	Disease incubation period in <i>Culex</i> mosquitoes	days	6		
$1/\gamma_2$	Infectiousness period in livestock	days	4		
q_1	Transovarial transmission rate in <i>Aedes</i> mosquitoes	-	0.05		
$1/\theta_1$	Development time of <i>Aedes</i> mosquitoes	days	10		
$1/\theta_3$	Development time of <i>Culex</i> mosquitoes	days	10		
K_1	Carrying capacity of <i>Aedes</i> mosquitoes	-	20,000	(Newton 1992)	
K_2	Carrying capacity of livestock	-	1,000		
K_3	Carrying capacity of <i>Culex</i> mosquitoes	-	20,000	(Newton 1992)	
VE	Vaccine efficacy	1/ day	Varies (See Tables 4 and 6)		
$v_l(t)$	Larvicide induced mortality	1/ day	Varies (See Table 5)		
$v_a(t)$	Adulticide induced mortality	1/ day	Varies (See Table 5)		
$V(t)$	Vaccination rate	1/ day	Varies (See Tables 4 and 6)		
$c(t)$	Culling rate	1/ day	Varies (See Table 7)		

Table 3. Initial conditions for the model of Rift Valley Fever transmission for all but preventive vaccination scenarios

Name	Symbol	Initial Value (General)
Uninfected <i>Aedes</i> eggs	P_1	5,000
Infected <i>Aedes</i> eggs	Q_1	0
Susceptible <i>Aedes</i>	S_1	4,999
Exposed <i>Aedes</i>	E_1	0
Infected <i>Aedes</i>	I_1	1
Susceptible livestock	S_2	900
Exposed livestock	E_2	0
Infected livestock	I_2	0
Removed livestock	R_2	0
Uninfected <i>Culex</i> eggs	P_3	5,000
Susceptible <i>Culex</i>	S_3	5,000
Exposed <i>Culex</i>	E_3	0
Infected <i>Culex</i>	I_3	0

*Table 4. Initial condition multiplicative factors for susceptible and immune livestock for the model of Rift Valley Fever transmission for preventive vaccination scenarios, where Immune = $PROP * 900$ and Susceptibles = $(1-PROP) * 900$*

Transmission Level	Target Population	Vaccine Efficacy	<i>PROP</i>
High	50%	65%	0.6648
High	75%	65%	0.7496
High	100%	65%	0.7999
Low	50%	85%	0.7227
Low	75%	85%	0.7967
Low	100%	85%	0.8395

Table 5. Simulated vector control scenarios

Intervention Type	Effectiveness	Timing	Trigger
Adulticide	30%	Periodic (4 weekly apps.)	Preventive
Larvicide	60%	One-Time (single app.)	Reactive
	90%	Constant (28 daily apps.)	

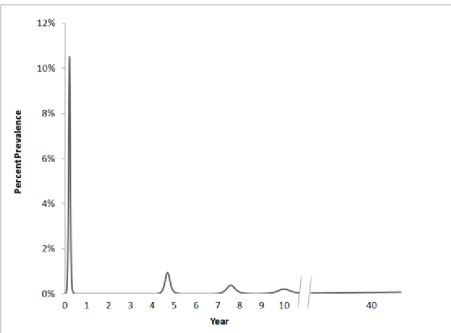
Table 6. Simulated livestock vaccination scenarios

Intervention Type	Effectiveness	Target Population	Campaign	Trigger
Vaccination	65%	50%	Periodic (4 weekly apps.)	Preventive
	85%	75%	One-Time (single app.)	Reactive - vector
		100%	Constant (28 daily apps.)	Reactive - livestock

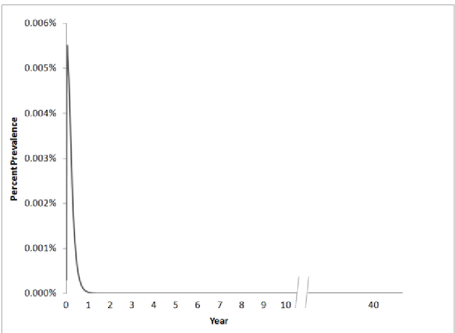
Table 7. Simulated livestock culling scenarios

Intervention Type	Compliance	Target Population	Timing	Trigger
Culling	50%	Exposed + Infected	Periodic	Reactive
	75%		One-Time	
	100%		Constant	

Figure 2. Percent livestock RVF prevalence for base-case of no intervention in an immunologically naive livestock population. The high transmission parameter set shows multiple minor outbreaks after the initial large outbreak of over 10% prevalence. The low transmission parameter set shows no subsequent outbreaks after the initial outbreak, which reaches 0.005% prevalence. (a) High transmission parameter set. (b) Low transmission parameter set.



a.



b.

removed state. Table 6 summarizes the livestock vaccination scenarios that have been simulated.

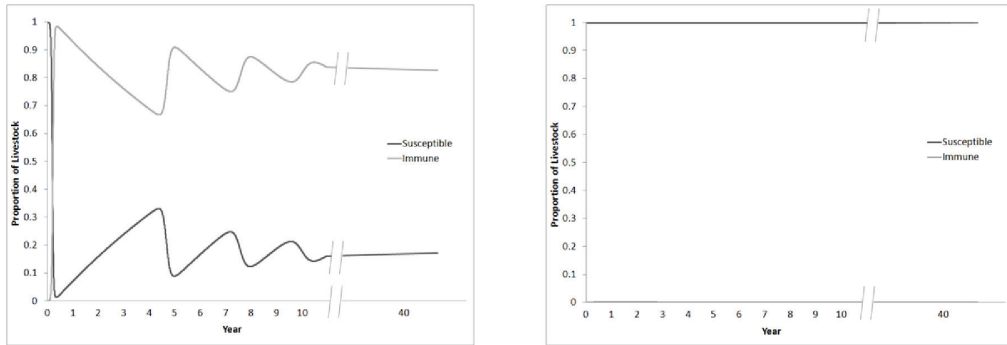
Livestock Culling Interventions

Livestock culling is incorporated into the model by removing exposed and infected animals from the cattle population according to a given compliance level (50%, 75% or 100%) when a RVF outbreak is detected in livestock, and thus

is employed on a reactive level only. Similar to vector control and livestock vaccination, culling occurs on a periodic, one-time or continuous basis (as defined previously). Table 7 summarizes the livestock culling scenarios that have been simulated.

All four types of interventions are triggered only once – either preventively at the beginning of the simulation or reactively in response to a detectable epidemic – and do not repeat with

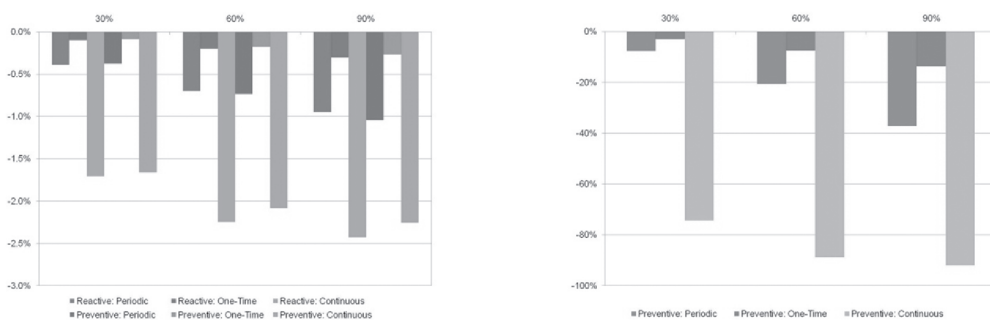
Figure 3. Proportion of susceptible and removed livestock for base-case of no intervention in an immunologically naïve livestock population. The high transmission set shows the cumulative effect of repeated outbreaks while the low transmission set reflects that nearly the entire live-stock population remains susceptible. (a) High transmission parameter set. (b) Low transmission parameter set.



a.

b.

Figure 4. Percent change in livestock deaths over base-case of no intervention for adulticide vector control scenarios with 30%, 60% and 90% effectiveness. It is important to note the scales for these two plots. The greatest reduction for the high transmission set is 2.5% while the low transmission set shows much larger reductions. This is a reflection of the low transmission set having a very small number of deaths, 0.2, for the base case. For both sets, the greatest reductions in deaths for all levels of effectiveness can be seen for the continuous application of adulticide. The low transmission sets never reached infection levels high enough to trigger reactive scenarios. Periodic and one-time applications showed dramatically less reductions. (a) High transmission parameter set. (b) Low transmission parameter set.



a.

b.

subsequent outbreaks that may occur. The one exception to this is livestock culling, in which we also simulated the impact of employing the removal of exposed and infected animals on an ongoing basis for the duration of the simulation.

RESULTS AND INTERPRETATION

In order to measure the effectiveness of the various intervention scenarios, outcomes for each in terms of cumulative livestock cases and deaths over the 10-year duration of the simulation were compared against the base-case of no intervention for both high and low transmission parameter sets. Under the base-case of no intervention, the 10-year cumulative numbers of livestock cases and deaths under high disease transmission are 1436 and 200, respectively; under low disease transmission, the cumulative number of livestock cases and deaths are 1.4 and 0.2, respectively. Percent prevalence and proportion of susceptible and removed livestock for high and low disease transmission are shown in Figures 2(a) and (b), and 3(a) and (b), respectively.

Impact of Vector Control

The percent change in livestock deaths due to RVF for the various vector control scenarios versus the base-case of no intervention are shown in Figures 4(a) and (b) and 5(a) and (b). Of the vector control scenarios simulated, when transmission of RVF is high, the greatest reduction in livestock deaths (approximately 2.5%) and cases is provided by the reactive application of adulticide for 28 consecutive days ("continuous" application) with 90% effectiveness. Note, however, that preventive vector control for this study implies that no cases or deaths have been detected in the livestock population prior to application of interventions, though mosquito infections and animal exposures have already begun to occur.

Under low disease transmission, the conditions to trigger reactive interventions fail to occur. Of the preventive vector control sce-

narios, however, the greatest impact on livestock deaths (greater than 90% reduction) and cases is again provided by preventive continuous adulticide application with 90% effectiveness, though when disease transmission is at these lower levels even under no intervention the predicted number of cases and deaths is nearly low enough to be negligible (1.4 and 0.2, respectively). As shown in the figures, the simulated application of larvicide at low transmission has the effect of increasing the number of livestock deaths. However, these changes are not as significant considering the nearly negligible number of cases and deaths in this case.

Impact of Livestock Vaccination

The percent change in livestock deaths due to RVF for the various livestock vaccination scenarios versus the base-case of no intervention are shown in Figures 6(a) and (b) and 7(a) and (b). Of the vaccination scenarios simulated, when transmission of RVF is high, the greatest reduction in livestock deaths (greater than 50% reduction) and cases is provided by the preventive immunization of 100% of the livestock population extended over 28 consecutive days ("continuous" vaccination) with 85% efficacy. For the case of preventive livestock vaccination, it has been assumed that a given number of animals have already received protection against RVF via immunization prior to the detection of disease in either vector or livestock populations. The vector- and livestock-reactive scenarios that impart the greatest benefit in terms of reduction in cases and deaths (approximately 8% and 5% reduction, respectively) are both also provided by continuous (28-day) vaccination of 100% of the population with 85% efficacy.

Under low disease transmission, the conditions to trigger livestock-reactive interventions fail to occur. However, of the preventive and vector-reactive vaccination scenarios, the greatest impact on livestock deaths (greater than 90% reduction) and cases is again provided by preventive vaccination of 100% of the population with 85% efficacy on a continuous basis for 28 days. The vector-reactive scenario which

Figure 5. Percent change in livestock deaths over base-case of no intervention for larvicide vector control scenarios with 30%, 60% and 90% effectiveness. The application of larvicide shows some more complex results. For the high transmission parameters, reactive continuous and periodic applications showed the greatest reduction in deaths. Proactive applications showed far lower benefits for the high transmission and actually amplified the low transmission set. This is reflective of the fact that fewer uninfected larvae are emerging to dilute the infected adults, and thus more infections are likely. Again, the low transmission base case predicted only 0.2 deaths so even small amplifications can appear dramatic. (a) High transmission parameter set. (b) Low transmission parameter set.

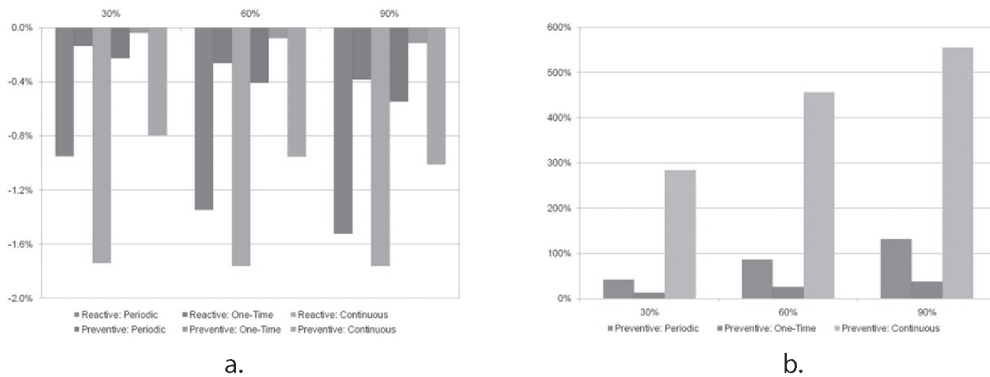


Figure 6. Percent change in livestock deaths over base-case of no intervention for livestock vaccination scenarios with 50%, 75% and 100% coverage and 65% vaccine efficacy. For both high and low transmission sets, any preventive application of vaccination has dramatically better results than the any reactive scenario. Also, it is important to note that in contrast to the previous figures, these reductions are 40% for the high transmission and over 80% for the low transmission. The reactive cases demonstrate that if one waits until a threshold percent of infected livestock or mosquitoes is met, it is too late to have such a dramatic reduction. The reductions of the reactive scenarios are on the order of the adulticide and larvicide scenarios. (a) High transmission parameter set. (b) Low transmission parameter set.

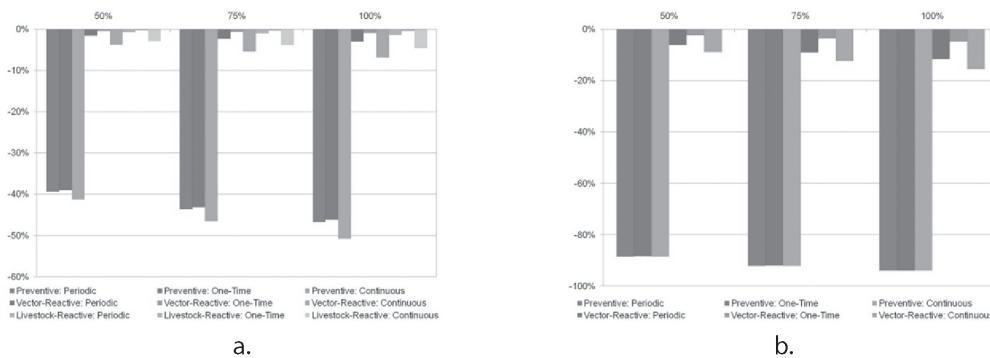
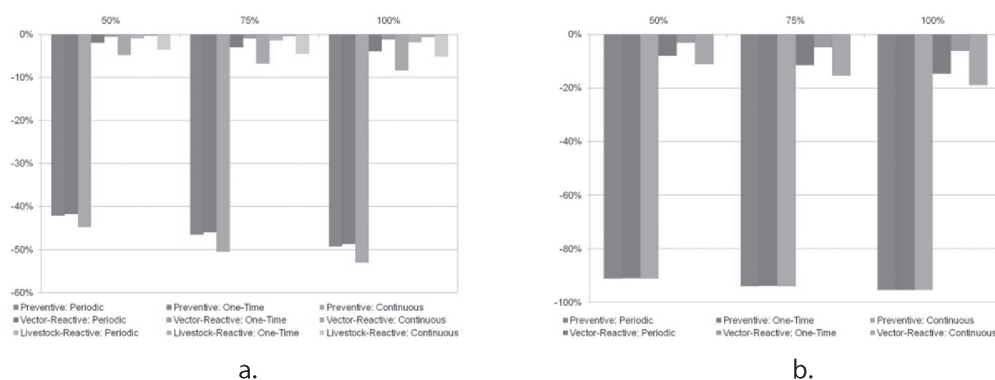


Figure 7. Percent change in livestock deaths over base-case of no intervention for livestock vaccination scenarios with 50%, 75% and 100% coverage and 85% vaccine efficacy. Increasing the efficacy of the vaccination is directly reflected in additional decreases in the livestock deaths. As with the lower efficacy, the best option is still preventive vaccination. (a) High transmission parameter set. (b) Low transmission parameter set.



imparts the greatest benefit in terms of reduction in cases and deaths (approximately 20% reduction) is also provided by continuous (28-day) vaccination of 100% of the population with 85% efficacy, keeping in mind that the total number of livestock cases and deaths even under the base-case of no intervention is still quite low.

Impact of Livestock Culling

The percent change in livestock deaths due to RVF for the various livestock culling scenarios versus the base-case of no intervention are shown in Figure 8. Of the culling scenarios simulated, when transmission of RVF is high, the greatest reduction in livestock deaths (greater than 15% reduction) and cases is provided by reactive culling of exposed and infected animals extended over 28 consecutive days (“continuous” culling) with 100% compliance. If such culling practices are continued for the duration of the simulated period (10 years) (“ongoing” culling), the drop in livestock deaths due to disease can be increased to almost 90%, with the bulk of animal infections occurring at the beginning of the simulation and the disease essentially disappearing from the system (without

outside introduction) after year 5 (Figure 9). Under low disease transmission, the conditions to trigger livestock-reactive interventions fail to occur and thus no culling scenarios are implemented under these conditions.

Parameter Sensitivity

To evaluate the sensitivity of the disease model to transmission parameters, we performed an exploratory analysis in which each of the four inter-species beta terms were varied individually over the range of possible values from a minimum of 0.05 to a maximum of 0.9 or 1.0, while the other three betas were held constant at their low transmission values. This was done in order to explore the degree of transmission at which a “detectable epidemic” (livestock prevalence of 1% or 2 livestock deaths in a given time-period) might occur and reactive interventions might be triggered.

In Figures 10(a) through (d), the total number of livestock cases for the duration of the simulation is plotted against beta values for each of the four livestock-reactive intervention types, plus the base-case of no intervention. The range of beta-conditions under which “detectable epidemic” thresholds are triggered (that is,

Figure 8. Percent change in livestock deaths over base-case of no intervention for livestock culling scenarios with 50%, 75% and 100% compliance. Results are shown only for the high transmission parameter set. All exposed and infected livestock are culled. Reactive, ongoing culling clearly has a significant impact on reducing the percent of livestock deaths from the disease, but dramatic levels of culling are required to achieve such results.

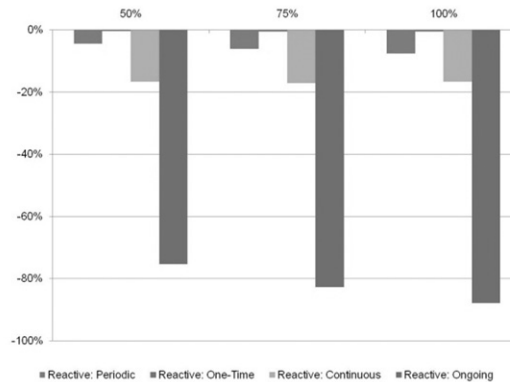


Figure 9. Disease state time-series plots for *Aedes* (top left), livestock (top right), and *Culex* (bottom left), and total livestock deaths and infections for ongoing livestock culling scenario with 100% compliance. All plots have days as the units for the x-axis and numbers of mosquitoes or livestock for the y-axis. The build-up of susceptible livestock can be seen in the upper right hand plot as the disease is clearly eliminated early in the scenario.

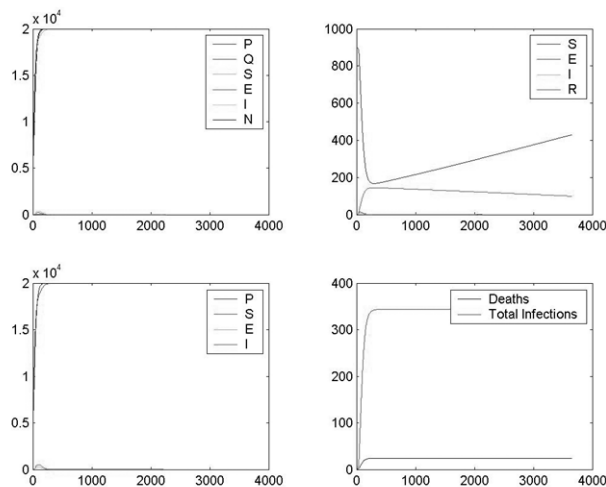
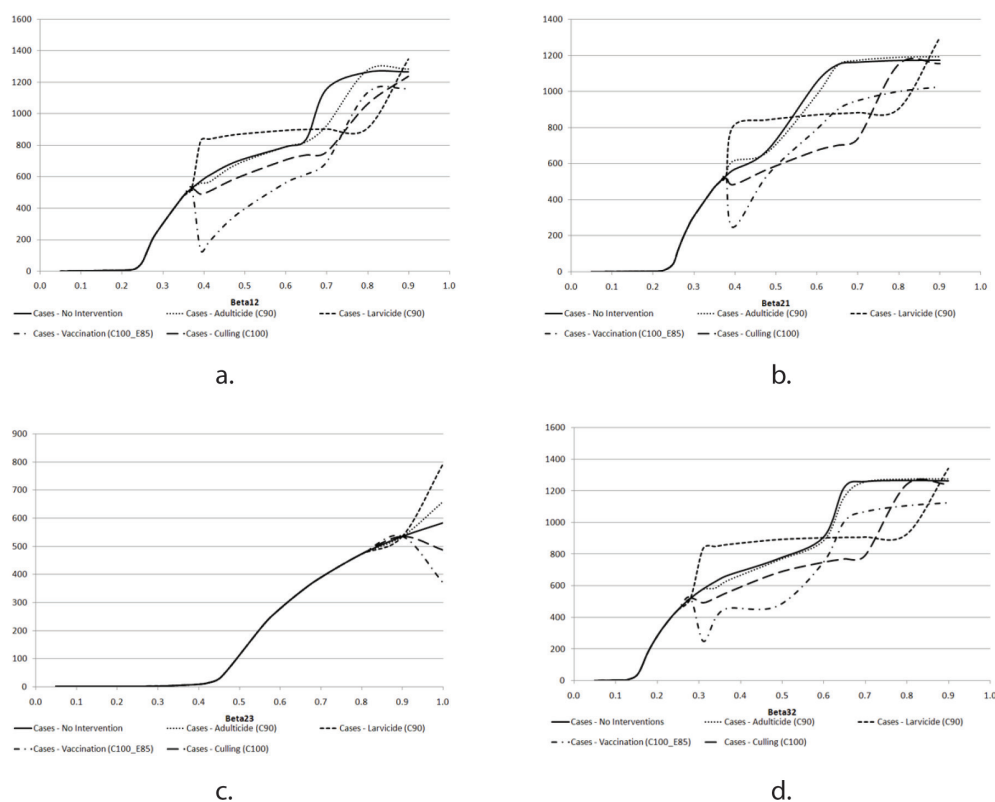


Figure 10. Total number of cases of RVF in livestock when individual transmission parameters are varied while the other three are held constant, under the base-case of no intervention as well as continuous application of adulticide (90% effectiveness), continuous application of larvicide (90% effectiveness), continuous livestock vaccination (100% coverage at 85% vaccine efficacy), and continuous livestock culling (100% compliance). As can be seen, the general trend for all parameters is an increase in the number of cases for an increase in the parameter; which is to be expected. The divergence for the various scenarios demonstrates the nonlinearities in the system. This also demonstrates that increased virulence could swamp even the best intervention strategy. (a) *Aedes* to livestock transmission. (b) Livestock to *Aedes* transmission. (c) Livestock to *Culex* transmission. (d) *Culex* to livestock transmission.



in a given time-step either 1% of the livestock population is infected or 2 livestock deaths occurred) occurs where variations between interventions are apparent on the graphs.

For each of the four betas explored there are definite tipping-points where cases and deaths increase abruptly and then level off, though for β_{23} (transmission from livestock to *Culex* mosquitoes) this happens at the upper end of the range. Within the range of beta val-

ues for which “detectable epidemics” occur there is a great deal of switching between which of the four intervention types minimizes total livestock cases, though continuous (28-day) vaccination of 100% of the livestock population at 85% efficacy and continuous livestock culling appear to perform best in general.

CONCLUSION

Main Simulation Results

At high disease transmission levels, all vector control strategies have a positive impact on reducing the 10-year cumulative number of livestock cases and deaths, with reactive application of adulticide offering the greatest reduction in livestock morbidity and mortality overall. However, even at the most effective of the scenarios simulated (reactive application of 90% effective adulticide for 28 consecutive days), adulticide prevents only 36 livestock cases and 5 deaths. Similar results are seen with the application of larvicide, with the most effective larvicide high transmission scenario preventing 26 livestock cases and 4 deaths. Thus, while the application of vector control is beneficial to livestock populations when transmission of Rift Valley Fever is high, the magnitude of this benefit must be compared to the cost associated with these interventions in order to evaluate their overall effectiveness. However, it is important to keep in mind that vector control interventions occur only once at the beginning of the simulation period, regardless of future outbreaks. Greater benefit to livestock may be seen if subsequent intervention pulses are simulated in response to later outbreaks in addition to the first.

Under conditions of low disease transmission, in which livestock morbidity and mortality is already low, adulticide prevents at most one case and has negligible effect on livestock deaths. The application of larvicide with low disease transmission, however, has the effect of *increasing* the number of cases and deaths in livestock as effectiveness and frequency of application increases. This may be associated with the fact that, since mosquito larvae are being proportionally reduced with each application, adult vector populations are no longer being diluted with un-infected newly emerged mosquitoes and as a result the density of infected adults increases with each larvicide application. The overall implication is that, under low disease transmission conditions, the application of

vector control strategies may not be sufficiently effective to justify the effort, and in the case of larvicide, may actually be contraindicated.

Regardless of disease transmission levels and vaccine efficacy, all livestock vaccination strategies have a positive effect on the reduction of morbidity and mortality in livestock populations. The greatest impact is provided by the preventive vaccination of 100% of livestock with 85% vaccine efficacy via a 28-day continuous campaign, which prevents 1,432 cases and all 200 deaths in the livestock population under high transmission, and prevents all 1.4 cases and 0.2 deaths under low transmission. This is, however, under the assumption that preventive vaccination occurs immediately prior to disease introduction in the system; subsequent delay between vaccination and disease introduction may decrease the degree of benefit as new un-vaccinated animals enter the population over time and dilute the level of herd immunity, since for the current study interventions are only implemented once regardless of future outbreaks.

Due to the nature of livestock culling interventions (killing all exposed and infected animals when detected), these scenarios were simulated only on a reactive level under high disease transmission, since detectable outbreak thresholds were not attained under low transmission conditions. Reactive livestock culling with 100% compliance for 28 consecutive days after the detection of an outbreak in the livestock population has the greatest impact on morbidity and mortality, preventing 189 livestock cases and 33 deaths over the duration of the 10-year simulation period, while 75% and 50% compliance offer similar benefits. Continuing a reactive culling strategy with 100% compliance beyond the initial 28 days through the end of the simulation, however, offers significantly greater benefits and overall has the effect of preventing almost 1,100 cases and 76 deaths. Under this strategy, Rift Valley Fever is eliminated from the three-species system after the final minor outbreaks occur in approximately year 5 of the simulation, even under high disease transmission.

Sensitivity Results

Model results point to a great deal of variability and switching in terms of which interventions offer the greatest benefit with respect to disease transmission levels. As can be seen in Figures 10(a), (b) and (d), over certain ranges of transmission parameter values, the application of larvicide performs the worst of all four intervention types, yet over a higher parameter range it becomes the best performer for a time, superseding even livestock vaccination which in general prevents a larger number of livestock cases than other interventions. However, in actual epidemic settings it is often difficult to pinpoint precise parameter values associated with the transmission of disease between species, and thus to discern which disease interventions might be optimal within that range. Looking at the range of transmission values in general, continuous livestock vaccination and continuous livestock culling appear to be the best choices in terms of intervention options as a whole.

There are a number of issues, which may have a significant impact on the choice of disease intervention, which have not been included in the current analysis. Associated with each intervention strategy is a series of costs, in terms of executing the interventions as well as the loss of livestock animals either to disease or culling, and in many cases these costs may outweigh the benefits derived to the point that the strategy should not be pursued. At significantly high costs, producers may be unwilling to comply with intervention strategies, regardless of potential benefit in terms of livestock morbidity and mortality reduction.

Additionally, as a vector-borne disease, Rift Valley Fever is driven in large part by environmental and geographic factors, which have not been incorporated into the current model. Factors such as temperature and precipitation may have significant impact on population dynamics in the *Aedes* and *Culex* vector populations, as well as disease dynamics, and may vary dramatically in a spatial context. Future

analysis is planned in which environmental parameters will be integrated into the model system and recommendations will be evaluated in a spatially heterogeneous setting. With these enhancements to the model optimal control techniques may be employed to offer policy recommendations on how best to respond to – and prevent – outbreaks of Rift Valley Fever in domestic livestock populations.

ACKNOWLEDGMENTS

The authors thank the thoughtful comments of the anonymous reviewers. DMH acknowledges support by the Research and Policy for Infectious Disease Dynamics (RAPIDD) program of the Science & Technology Directorate, Department of Homeland Security and Fogarty International Center, National Institutes of Health. Funding from the DHS Center of Excellence for Foreign Animal and Zoonotic Disease Defense (FAZD Center) is also acknowledged.

REFERENCES

- Centers for Disease Control and Prevention (CDC). (1998). Rift Valley Fever-East Africa, 1997-1998. *Morbidity and Mortality Weekly Report*, 47(13), 261–264.
- Daubney, R., Hudson, J. R., & Garnham, P. C. (1931). Enzootic hepatitis of Rift Valley Fever, an undetectable virus disease of sheep, cattle and man from East Africa. *The Journal of Pathology and Bacteriology*, 34, 543–579. doi:10.1002/path.1700340418
- Gaff, H. D., Hartley, D. M., & Leahy, N. P. (2007). An epidemiological model of Rift Valley Fever. *Electronic Journal of Differential Equations*, 115, 1–12.
- Gerdes, G. H. (2004). Rift Valley Fever. *Revue Scientifique et Technique (International Office of Epizootics)*, 23(2), 613–623.
- Jupp, P. G., Kemp, A., Grobbelaar, A., Leman, P., Burt, F. J., & Alahmed, A. M. (2002). The 2000 epidemic of Rift Valley Fever in Saudi Arabia: Mosquito vector studies. *Medical and Veterinary Entomology*, 15(3), 245–252. doi:10.1046/j.1365-2915.2002.00371.x

Newton, E. A. C., & Reiter, P. (1992). A model of the transmission of dengue fever with an evaluation of the impact of ultra-low volume (ULV) insecticide applications on dengue epidemics. *The American Journal of Tropical Medicine and Hygiene*, 47, 709–720.

Peters, C. J., & Linthicum, K. J. (1994). Rift Valley Fever. In G. W. Beran (Ed.), *Handbook of zoonoses, section B: Viral zoonoses* (2nd ed.) (pp. 125-138). Boca Raton, FL: CRC Press.

Turell, M. J., Dohm, D. J., Mores, C. N., Terracina, L., Walette, D. L., & Hribar, L. J. (2008). Potential for North American mosquitoes to transmit Rift Valley Fever virus. *Journal of the American Mosquito Control Association*, 24(4), 502–507. doi:10.2987/08-5791.1

World Health Organization (WHO). (2007). *Rift Valley Fever*. Retrieved from <http://www.who.int/mediacentre/factsheets/fs207/en/>

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